JavaScript is disabled on your browser.

compact1, compact2, compact3

java.nio.file

Class FileStore

java.lang.Object
    java.nio.file.FileStore

```
public abstract class FileStore
extends Object
```

Storage for files. A `FileStore` represents a storage pool, device, partition, volume, concrete file system or other implementation specific means of file storage. The `FileStore` for where a file is stored is obtained by invoking the `getFileStore` method, or all file stores can be enumerated by invoking the `getFileStores` method.

In addition to the methods defined by this class, a file store may support one or more `FileStoreAttributeView` classes that provide a read-only or updatable view of a set of file store attributes.

Since:

1.7

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	**FileStore**()	Initializes a new instance of this class.

Method Summary

Abstract Methods

Modifier and Type	Method
abstract **Object**	**getAttribute**(**String** attribute)
abstract <V extends **FileStoreAttributeView**> V	**getFileStoreAttributeView**(**Class**<V> t
abstract long	**getTotalSpace**()
abstract long	**getUnallocatedSpace**()
abstract long	**getUsableSpace**()
abstract boolean	**isReadOnly**()
abstract **String**	**name**()
abstract boolean	**supportsFileAttributeView**(**Class**<? extends **FileAttributeView**> type)
abstract boolean	**supportsFileAttributeView**(**String** nam
abstract **String**	**type**()

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

FileStore

```
protected FileStore()
```

Initializes a new instance of this class.

Method Detail

name

```
public abstract String name()
```

Returns the name of this file store. The format of the name is highly implementation specific. It will typically be the name of the storage pool or volume.

The string returned by this method may differ from the string returned by the `toString` method.

Returns:

the name of this file store

type

```
public abstract String type()
```

Returns the *type* of this file store. The format of the string returned by this method is highly implementation specific. It may indicate, for example, the format used or if the file store is local or remote.

Returns:

a string representing the type of this file store

isReadOnly

```
public abstract boolean isReadOnly()
```

Tells whether this file store is read-only. A file store is read-only if it does not support write operations or other changes to files. Any attempt to create a file, open an existing file for writing etc. causes an `IOException` to be thrown.

Returns:

`true if, and only if, this file store is read-only`

getTotalSpace

```
public abstract long getTotalSpace()
                            throws IOException
```

Returns the size, in bytes, of the file store.

Returns:

`the size of the file store, in bytes`

Throws:

`IOException - if an I/O error occurs`

getUsableSpace

```
public abstract long getUsableSpace()
                             throws IOException
```

Returns the number of bytes available to this Java virtual machine on the file store.

The returned number of available bytes is a hint, but not a guarantee, that it is possible to use most or any of these bytes. The number of usable bytes is most likely to be accurate immediately after the space attributes are obtained. It is likely to be made inaccurate by any external I/O operations including those made on the system outside of this Java virtual machine.

Returns:

`the number of bytes available`

Throws:

`IOException - if an I/O error occurs`

getUnallocatedSpace

```
public abstract long getUnallocatedSpace()
                                    throws IOException
```

Returns the number of unallocated bytes in the file store.

The returned number of unallocated bytes is a hint, but not a guarantee, that it is possible to use most or any of these bytes. The number of unallocated bytes is most likely to be accurate immediately after the space attributes are obtained. It is likely to be made inaccurate by any external I/O operations including those made on the system outside of this virtual machine.

Returns:

the number of unallocated bytes

Throws:

IOException - if an I/O error occurs

supportsFileAttributeView

```
public abstract boolean supportsFileAttributeView(Class<? extends FileAttributeVi
```

Tells whether or not this file store supports the file attributes identified by the given file attribute view.

Invoking this method to test if the file store supports BasicFileAttributeView will always return true. In the case of the default provider, this method cannot guarantee to give the correct result when the file store is not a local storage device. The reasons for this are implementation specific and therefore unspecified.

Parameters:

type - the file attribute view type

Returns:

true if, and only if, the file attribute view is supported

supportsFileAttributeView

```
public abstract boolean supportsFileAttributeView(String name)
```

Tells whether or not this file store supports the file attributes identified by the given file attribute view.

Invoking this method to test if the file store supports BasicFileAttributeView, identified by the name "basic" will always return true. In the case of the default provider, this method cannot guarantee to

give the correct result when the file store is not a local storage device. The reasons for this are implementation specific and therefore unspecified.

Parameters:

`name - the name of file attribute view`

Returns:

`true if, and only if, the file attribute view is supported`

getFileStoreAttributeView

```
public abstract <V extends FileStoreAttributeView> V getFileStoreAttributeView(Cl
```

Returns a `FileStoreAttributeView` of the given type.

This method is intended to be used where the file store attribute view defines type-safe methods to read or update the file store attributes. The `type` parameter is the type of the attribute view required and the method returns an instance of that type if supported.

Type Parameters:

`V - The FileStoreAttributeView type`

Parameters:

`type - the Class object corresponding to the attribute view`

Returns:

`a file store attribute view of the specified type or null if the attribute view is not available`

getAttribute

```
public abstract Object getAttribute(String attribute)
                             throws IOException
```

Reads the value of a file store attribute.

The `attribute` parameter identifies the attribute to be read and takes the form:

view-name`:`*attribute-name*

where the character `':'` stands for itself.

view-name is the `name` of a `AttributeView` that identifies a set of file attributes. *attribute-name* is the name of the attribute.

Usage Example: Suppose we want to know if ZFS compression is enabled (assuming the "zfs" view is supported):

```
    boolean compression = (Boolean)fs.getAttribute("zfs:compression");
```

Parameters:

`attribute` - the attribute to read

Returns:

the attribute value; null may be a valid valid for some attributes

Throws:

`UnsupportedOperationException` - if the attribute view is not available or it does not support reading the attribute

`IOException` - if an I/O error occurs

Submit a bug or feature

For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.

Copyright © 1993, 2026, Oracle and/or its affiliates. All rights reserved. Use is subject to license terms. Also see the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.